1.1.1. Quick Sort 29:23

Write a program to sort an array of length n using the Quick Sort algorithm. The program should read the array elements and print the array before and after sorting.

Input Format:

- The first line contains a single integer n, representing the number of elements in the array.
- The second line contains n space-separated integers representing the elements of the array.

Output Format:

- The first line of output contains space-separated integers representing the elements of the original array
- The Second input line contains space-separated integers representing the elements after sorting.

Constraints:

- $1 \leq n \leq 1000$

Note:

- The driver code is already provided. You need to complete the required code to perform Quick Sort correctly.

Sample Test Cases +

Explorer

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Debugger

```c
#include <stdio.h>
void swap(int *a, int *b){
    int temp=*a;
    *a=*b;
    *b=temp;
    //SHOURYAM GARG 25070521094
}
 int partition(int arr[],int low, int high) {
     int pivot= arr[high];
     int i = low-1;
for(int j=low;j<high;j++){
    if(arr[j]<pivot){
        i++;

swap(&arr[i],&arr[j]);
    }
}

swap(&arr[i+1],&arr[high]);
     return(i+1);
 }
// Quick Sort function
void quickSort(int arr[], int low, int high) {
    //SHOURYAM GARG 25070521094
    if(low<high){
        int pi= partition(arr,low,high);

quickSort(arr,low,pi-1);

quickSort(arr,pi+1,high)
```

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```
    }

    //SHOURYAM GARG 25070521094
}

// Function to print an array
void printArray(int arr[], int size) {
    for (int i = 0; i < size; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}

int main() {
    int n;
    // Input the size of the array
    scanf("%d", &n);
    int arr[n];

    // Input the elements of the array
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // Print the original array
    printArray(arr, n);

    // Sort the array using quick sort
    quickSort(arr, 0, n - 1);

    // Print the sorted
```

```
58      printArray(arr, n);
59
60      return 0;
61  } //SHOURYAM GARG
    25070521094
```